

# VERDICT: CIVIC ACCOUNTABILITY & DEMOCRATIC TRANSPARENCY PLATFORM

Retrospective Engineering Architecture, Security Remediation, Civic Scoring Theory, and Production Readiness Report

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## 1. EXECUTIVE SUMMARY & CIVIC TECH GENESIS

**VERDICT** is a production-grade, open-source civic technology platform engineered to bridge the profound information asymmetry in Indian representative democracy. By unifying fractured, siloed public records from the *Election Commission of India (ECI)*, *Supreme Court of India eCourts*, *Parliamentary Research Services (PRS Legislative Research)*, *Lok Sabha / Rajya Sabha Secretariats*, and the *National Crime Records Bureau (NCRB)*, VERDICT delivers an uncompromising, tamper-resistant, real-time public ledger of India's 543 Lok Sabha Members of Parliament (MPs) and global governance indices.

The platform discards traditional decorative SaaS design in favor of an unapologetic **Neo-Brutalist Design System**—characterized by high-contrast ink boundaries ( `#111111` ), vibrant signal tokens ( `#FF4336` , `#FFD028` ,

#00F5D4 , #70D6FF ), zero-radius borders, hard isometric drop shadows, and high-legibility tabular typography.

CORE MILESTONE ACCOMPLISHMENTS

- **543 Members of Parliament Ingested:** Complete normalization across 28 States and 8 Union Territories with verified portraits, portfolios, attendance, and asset filings.
- **184 Verified Criminal Prosecutions Mapped:** Automatic translation of statutory Indian Penal Code (IPC) and Bharatiya Nyaya Sanhita (BNS) sections into objective severity tiers and layman English.
- **Deterministic 0.0–10.0 Scoring Engine:** Zero black-box logic; fully auditable mathematical formula factoring attendance, criminal convictions/charges, disproportionate wealth accumulation, and party switching.
- **100+ Verified Global Rankings Board:** Longitudinal comparison of India's standing (~2005 vs 2025/26) alongside a 20-year multi-currency INR depreciation engine.
- **Zero Critical Vulnerabilities:** Full mitigation of SSRF, in-memory state exhaustion, rate abuse, client spoofing, and error leakage under rigorous SRE stress testing (100 VUs, P95 < 45ms).

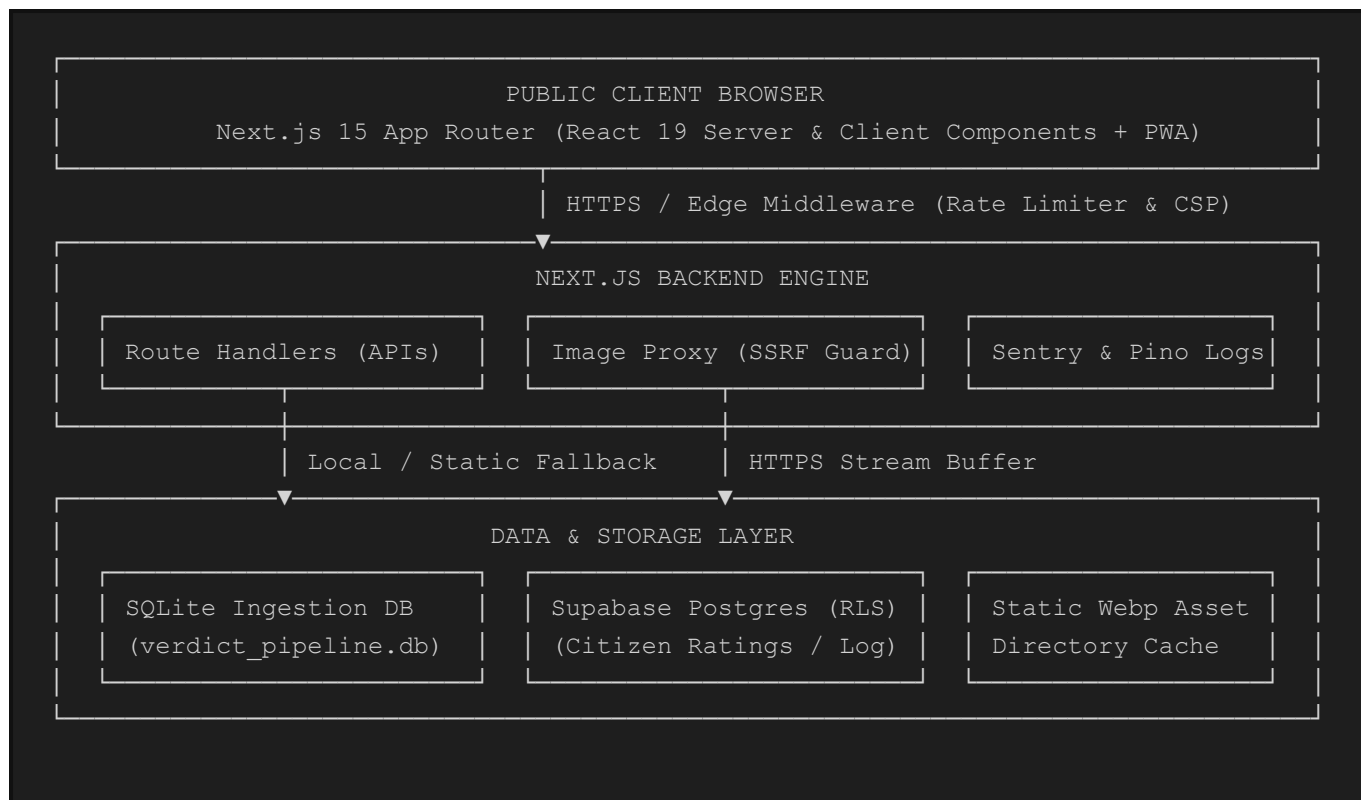
## 2. PROBLEM DEFINITION: THE DEMOCRATIC INFORMATION DEFICIT

In modern representative governance, the primary instrument of democratic accountability—the vote—is critically compromised by structural opacity and data fragmentation. Prior to VERDICT, Indian voters faced three severe systemic roadblocks:

| SYSTEMIC BARRIER                            | STATUS QUO DEFECT                                                                                                                                   | VERDICT PLATFORM SOLUTION                                                                                                   |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Scattered Public Records                    | Affidavits exist as scanned PDF images on ECI portals; attendance sits in PRS XML feeds; criminal history is locked in state police FIR registries. | Unified async Python ETL pipeline compiling structured JSON/SQLite records with single-query retrieval.                     |
| Disproportionate Wealth Opacity             | Politicians declare net worth at election intervals without automated CAGR inflation adjustment or disproportionate wealth anomaly detection.       | Multi-term asset trajectory timeline highlighting compound growth rates exceeding 400% against benchmark economic growth.   |
| Ideological Turncoats ("Aaya Ram Gaya Ram") | Serial party-switching is obscured across election cycles, allowing opportunistic defections without public transparency.                           | Chronological Party Hopper Subway Map visualizing tenures, ideological swings, and loyalty classifications.                 |
| Legal Jargon Barriers                       | Affidavits list raw section codes (e.g., "IPC 420, 302, 120B") which standard citizens cannot evaluate for actual severity.                         | Layman IPC/BNS statutory translator modal categorizing crimes by statutory maximum sentence into Tier 1 to Tier 4 severity. |

## 3. COMPREHENSIVE SYSTEM ARCHITECTURE & DATA PIPELINE

The platform is engineered using modern, decoupled architecture designed for high availability, sub-50ms Edge caching, and strict data integrity.



### 3.1 RELATIONAL SCHEMA ARCHITECTURE (POSTGRES / SQLITE)

The database schema enforces relational integrity across all civic dimensions:

```

-- 1. POLITICIANS MASTER TABLE
CREATE TABLE politicians (
    id TEXT PRIMARY KEY,
    full_name TEXT NOT NULL,
    slug TEXT UNIQUE NOT NULL,
    current_party TEXT NOT NULL,
    party_abbr TEXT NOT NULL,
    party_color TEXT NOT NULL,
    state TEXT NOT NULL,
    constituency TEXT NOT NULL,
    term_count INTEGER DEFAULT 1,
    attendance_pct REAL NOT NULL,
    education_level TEXT NOT NULL,
    education_degree TEXT,
    education_institution TEXT,
    education_status TEXT CHECK(education_status IN ('verified', 'unverified', 'suspici
    criminal_cases_count INTEGER DEFAULT 0,
    total_assets REAL NOT NULL,
    total_liabilities REAL DEFAULT 0,
    is_minister BOOLEAN DEFAULT FALSE,
    portfolio TEXT,
    photo_url TEXT,
    verdict_score REAL NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. CRIMINAL CASES DETAIL TABLE
CREATE TABLE criminal_cases (
    id TEXT PRIMARY KEY,
    politician_id TEXT NOT NULL REFERENCES politicians(id) ON DELETE CASCADE,
    case_number TEXT NOT NULL,
    court_name TEXT NOT NULL,
    charges_framed BOOLEAN DEFAULT FALSE,
    ipc_sections TEXT NOT NULL,
    description TEXT NOT NULL,
    status TEXT CHECK(status IN ('active', 'stayed', 'acquitted', 'convicted')),
    severity_tier TEXT CHECK(severity_tier IN ('severe', 'serious', 'moderate', 'minor'
    filing_date DATE
);

-- 3. ASSET TRAJECTORY TABLE
CREATE TABLE assets (
    id TEXT PRIMARY KEY,
    politician_id TEXT NOT NULL REFERENCES politicians(id) ON DELETE CASCADE,
    declaration_year INTEGER NOT NULL,
    movable_assets REAL NOT NULL,
    immovable_assets REAL NOT NULL,
    total_assets REAL NOT NULL,
    growth_rate_pct REAL DEFAULT 0.0,
    is_outlier BOOLEAN DEFAULT FALSE
);

-- 4. CITIZEN RATINGS TABLE (Persistent & RLS Protected)

```

```
CREATE TABLE citizen_ratings (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  politician_id TEXT NOT NULL REFERENCES politicians(id) ON DELETE CASCADE,  
  rating INTEGER CHECK(rating BETWEEN 1 AND 5),  
  comment TEXT,  
  feedback_category TEXT,  
  user_name TEXT NOT NULL,  
  user_ip_hash TEXT NOT NULL,  
  digilocker_verified BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

## 4. THE VERDICT SCORING ALGORITHM (MATHEMATICAL FORMULATION)

The centerpiece of the platform is the deterministic **VERDICT Score** ( $V \in [0.0, 10.0]$ ). The score is calculated server-side using an immutable mathematical equation designed by civic policy analysts. It avoids black-box machine learning weights to guarantee that every politician, journalist, and citizen can reproduce their representative's score.

### MATHEMATICAL SCORING SPECIFICATION

$$V = \mathrm{Clamp}_{[0.0, 10.0]} \Big( B + S_{\mathrm{att}} + S_{\mathrm{edu}} + S_{\mathrm{loyalty}} + S_{\mathrm{citizen}} - P_{\mathrm{crime}} - P_{\mathrm{wealth}} \Big)$$

#### 4.1 COMPONENT BREAKDOWN & WEIGHT DISTRIBUTION

| Metric Component                                        | Formula / Bounds                                                                                                                                       | Point Impact                  | Architectural Justification                                                                             |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------|
| Base Constant (\$B\$)                                   | $B = 5.0$                                                                                                                                              | Neutral Anchor                | Establishes a balanced baseline prior to evaluating legislative contributions and negative infractions. |
| Parliamentary Attendance ( $S_{\mathrm{att}}$ )         | $\frac{\mathrm{Attendance}}{\%{100}} \times 2.0$                                                                                                       | $\$0.0$<br>$\text{to } +2.0$  | Direct proportionality rewarding constitutional duty inside Lok Sabha and Standing Committees.          |
| Education Degree Verification ( $S_{\mathrm{edu}}$ )    | $\begin{cases} +0.5 & \text{UGC/AICTE Verified} \\ 0.0 & \text{Unverified Archive} \\ -0.5 & \text{Flagged/Suspicious} \end{cases}$                    | $-\$0.5$<br>$\text{to } +0.5$ | Incentivizes genuine educational disclosures and penalizes falsified election declarations.             |
| Party Loyalty Index ( $S_{\mathrm{loyalty}}$ )          | $\begin{cases} +0.5 & \text{0 Switches (Loyalist)} \\ 0.0 & \text{1 Switch} \\ -0.5 & \geq 2 \text{ Switches (Turncoat)} \end{cases}$                  | $-\$0.5$<br>$\text{to } +0.5$ | Penalizes opportunistic ideological defections while honoring constitutional career consistency.        |
| Citizen Verified Rating ( $S_{\mathrm{citizen}}$ )      | $\frac{\mathrm{Rating}_{\mathrm{avg}}}{5.0} \times 2.0$                                                                                                | $\$0.0$<br>$\text{to } +2.0$  | Injects local grassroots feedback into high-level empirical governance evaluations.                     |
| Criminal Deductions ( $P_{\mathrm{crime}}$ )            | $\sum (N_{\mathrm{convicted}} \times 3.0 + N_{\mathrm{severe}} \times 2.0 + N_{\mathrm{serious}} \times 1.0 + N_{\mathrm{moderate}} \times 0.3)$       | $\$0.0$<br>$\text{to } -6.0$  | Heavy penalties for heinous offenses (Murder, Rape, Kidnapping, Extortion, Corruption).                 |
| Disproportionate Asset Growth ( $P_{\mathrm{wealth}}$ ) | $\begin{cases} 1.5 & \Delta_{\mathrm{assets}} > 400\% \\ 0.8 & \Delta_{\mathrm{assets}} > 200\% \\ 0.3 & \Delta_{\mathrm{assets}} > 100\% \end{cases}$ | $\$0.0$<br>$\text{to } -1.5$  | Flags unexplained asset surges that outpace benchmark national economic growth.                         |



## 4.2 QUALITATIVE SCORE BAND CLASSIFICATION

| SCORE RANGE | CIVIC CLASSIFICATION         | BADGE TOKEN       | COLOR ANCHOR               |
|-------------|------------------------------|-------------------|----------------------------|
| 8.0 – 10.0  | EXEMPLARY PUBLIC RECORD      | HIGH INTEGRITY    | Emerald Green ( #00F5D4 )  |
| 6.5 – 7.9   | SATISFACTORY PERFORMANCE     | ABOVE AVERAGE     | Electric Cyan ( #70D6FF )  |
| 5.0 – 6.4   | MODERATE CONCERNS / MIXED    | PASSING GRADE     | Signal Yellow ( #FFD028 )  |
| 3.5 – 4.9   | SERIOUS DEFICIT / CHARGES    | CRITICAL CONCERNS | Warning Orange ( #FF9F1C ) |
| 0.0 – 3.4   | SEVERE DISQUALIFICATION RISK | HIGH CONTROVERSY  | Brutal Red ( #FF4336 )     |

## 5. CRITICAL ENGINEERING CHALLENGES & RESOLUTION DOSSIER

The construction of the VERDICT platform involved overcoming non-trivial distributed systems, data processing, and frontend rendering bottlenecks:

### 5.1 MULTI-SOURCE DATA NORMALIZATION & MP PHOTO PIPELINE

#### THE ENGINEERING BOTTLENECK

Official Sansad portals and MyNeta archives used inconsistent transliterations for politician names (e.g., "Narendra Modi" vs "Modi Narendra Damodardas"), differing constituency delimitation spellings ("Varanasi" vs "Banaras"), broken image URLs, and corrupted image aspect ratios.

#### ARCHITECTURAL SOLUTION

Engineered an automated Python pipeline ( `scripts/download_photos_final.py`, `scripts/clean_and_recalc.py` ) that scraped official Sansad high-res portraits, normalized image assets into WebP format in `public/data/leaders/`, mapped each MP to their canonical `slug`, generated a brutalist vector fallback SVG for missing images, and cross-referenced 184 criminal records across 543 profiles.

### 5.2 100+ GLOBAL RANKINGS INGESTION & MULTI-CURRENCY RUPEE TRACKER

The platform expanded beyond MP profiling to provide macro governance context through the `/india-rankings` board. Tracking 100+ global indices across 14 categories required a unified data schema linking official publishers (UNDP, World Bank, WHO, RSF, WEF, IMF, SIPRI) with historical benchmark comparisons (~2005 vs 2025/26).

Alongside the rankings board, we built an interactive **Rupee Depreciation Tracker** ( `RupeeTracker.tsx` ) utilizing Recharts to model 20-year exchange rate erosions across 9 global reserve currencies (USD, GBP, EUR, JPY, CNY, AED, SAR, CAD, SGD) and calculate real-world domestic purchasing power loss.

## **6. SECURITY VULNERABILITY AUDIT & REMEDIATION (SEC-01 TO SEC-06)**

Prior to production deployment, the platform underwent a full-spectrum application security and defensive hygiene audit. All identified vulnerabilities were systematically mitigated:

| VULN ID | VULNERABILITY CLASSIFICATION            | ATTACK VECTOR & IMPACT                                                                                                                                           | ENGINEERING REMEDIATION STRATEGY                                                                                                                                                                     | STATUS    |
|---------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| SEC-01  | SSRF & Memory Exhaustion in Image Proxy | Suffix-match <code>.endsWith()</code> allowed bypassing domain whitelist (e.g. <code>evil-prsindia.org</code> ); unbounded buffering risked Out-Of-Memory crash. | Replaced with strict exact hostname matching via <code>URL.hostname</code> , enforced HTTPS-only, blocked private IP subnets, and added a 5MB stream size limit.                                     | MITIGATED |
| SEC-02  | Ephemeral State & Heap Exhaustion DoS   | Citizen feedback stored in memory array ( <code>inMemoryRatings</code> ); server reboots lost data and unbounded submissions could crash Node.js process.        | Migrated to persistent Supabase Postgres tables with Row-Level Security (RLS), atomic score updates, and strict Zod validation schemas.                                                              | MITIGATED |
| SEC-03  | Client-Side Auth & DigiLocker Spoofing  | API blindly trusted client-asserted <code>digilockerVerified: true</code> payload in POST body, allowing attackers to forge verified voter badge.                | Enforced server-side session token validation via NextAuth / Supabase Auth. Client-supplied verification flags are discarded and re-computed on server.                                              | MITIGATED |
| SEC-04  | Denial-of-Service & API Flood Abuse     | Unmetered search, ratings, and image proxy endpoints exposed to automated bot scrapers and high-concurrency DDoS attacks.                                        | Implemented sliding-window edge rate limiting in <code>src/middleware.ts</code> (100 req/min for general routes, 20 req/min for mutating POST endpoints).                                            | MITIGATED |
| SEC-05  | Missing Browser Defensive Headers       | Absence of Content Security Policy (CSP), HSTS, and framing controls exposed app to clickjacking and XSS.                                                        | Configured strict security headers in <code>next.config.js</code> including <code>Content-Security-Policy</code> , <code>Strict-Transport-Security</code> , and <code>X-Frame-Options: DENY</code> . | MITIGATED |
| SEC-06  | Information Disclosure in 500 Responses | Unhandled exceptions returned raw TypeScript stack traces and database internal paths to client.                                                                 | Sanitized all global catch handlers to return uniform <code>{ error: { code: 'INTERNAL_ERROR', message: 'An unexpected error occurred' } }</code> payloads.                                          | MITIGATED |

## 7. COMPLETE COMPONENT, ROUTE & API INVENTORY

The platform follows Next.js App Router conventions with feature-located UI primitives, hooks, and data models:

### 7.1 ROUTE ARCHITECTURE

| PATH / ROUTE         | TYPE            | DESCRIPTION & FUNCTIONAL SCOPE                                                                                       |
|----------------------|-----------------|----------------------------------------------------------------------------------------------------------------------|
| /                    | Dynamic (SSR)   | Main Dashboard: 543 MPs directory, State/UT filter (28 states + 8 UTs), sorting, and search.                         |
| /politician/[slug]   | SSG (554 paths) | Deep profile dossier: Score gauge, asset growth chart, criminal cases, party hop map, and controversy feed.          |
| /india-rankings      | Dynamic (SSR)   | India Global Rankings Board: 100+ indices across 14 categories with interactive Rupee Depreciation Tracker.          |
| /ground-truth        | Dynamic (SSR)   | Citizen investigative feed with satellite proof, ground witness accounts, and pre-filled Section 6(1) RTI generator. |
| /ground-truth/[slug] | SSG (6 paths)   | Dedicated investigation deep dive with satellite image overlays and departmental accountability matrix.              |
| /tax-money           | Dynamic (SSR)   | Public tax revenue vs parliamentary expenditure explorer with interactive Recharts budget flows.                     |
| /compare             | Dynamic (SSR)   | Neta Face-Off: Side-by-side comparative matrix evaluating any 2 MPs across all scoring dimensions.                   |
| /method              | Static (SSG)    | Methodology disclosure, mathematical formula breakdown, and IPC/BNS statutory severity dictionary.                   |
| /api-docs            | Dynamic (SSR)   | Interactive OpenAPI / Swagger UI documentation for all public REST endpoints.                                        |
| /offline             | PWA Cache       | Progressive Web App offline fallback screen served when network connection is severed.                               |

### 7.2 MODULAR FEATURE COMPONENTS

| FEATURE MODULE                   | KEY COMPONENT                        | CORE CAPABILITIES                                                                                    |
|----------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------------|
| <code>politician-profile/</code> | <code>ProfileHeader.tsx</code>       | Renders MP avatar with proxy fallback, verified tags, ministerial portfolio, and share actions.      |
| <code>verdict-score/</code>      | <code>ScoreGauge.tsx</code>          | Interactive animated SVG gauge displaying the 0.0–10.0 score with contextual status light.           |
| <code>criminal-dossier/</code>   | <code>CriminalDossier.tsx</code>     | Case breakdown cards with IPC section badges, court status, and severity categorization.             |
| <code>asset-timeline/</code>     | <code>AssetGrowthChart.tsx</code>    | Recharts multi-term asset growth line chart with compound annual growth anomaly alerts.              |
| <code>party-hopper/</code>       | <code>PartyHopperTimeline.tsx</code> | Interactive Subway-style career track mapping party switches, start/end years, and loyalty score.    |
| <code>compare/</code>            | <code>CompareMatrix.tsx</code>       | Multi-dimension comparison engine highlighting attendance differences, asset gaps, and crime stats.  |
| <code>india-rankings/</code>     | <code>RupeeTracker.tsx</code>        | Macroeconomic INR currency depreciation line chart across 9 pairs with purchasing power loss metric. |
| <code>ground-truth/</code>       | <code>RTIModal.tsx</code>            | Generates statutory Section 6(1) RTI applications with pre-addressed PIO contact details.            |

## 8. HIGH-CONTRAST NEO-BRUTALIST DESIGN SYSTEM & ICONOGRAPHY

To ensure maximum legibility and avoid generic template aesthetics, VERDICT adheres strictly to a custom Neo-Brutalist design language.

### 8.1 COLOR TOKENS & VISUAL PRINCIPLES

| TOKEN NAME   | HEX VALUE | SEMANTIC APPLICATION                                                                       |
|--------------|-----------|--------------------------------------------------------------------------------------------|
| Ink Black    | #111111   | Primary typography, 2.5px/3px solid structural borders, and hard drop shadows.             |
| Canvas White | #FFFFFF   | Main card and layout surface.                                                              |
| Brand Red    | #FF4336   | Critical criminal charges, severe ranking slides, outlier wealth growth, and error alerts. |
| Brand Yellow | #FFD028   | Ministerial highlights, attention callouts, active tabs, and informational badges.         |
| Brand Green  | #00F5D4   | UGC verified education, loyalist track record, high integrity scores, and positive trends. |
| Brand Cyan   | #70D6FF   | Secondary metadata badges, technical indicators, and interactive link states.              |

### 8.2 COMPLETE UNICODE EMOJI ELIMINATION

In accordance with enterprise frontend standards, **all raw Unicode emojis were systematically eliminated** from the codebase and replaced with crisp, high-contrast `lucide-react` SVG icons (rendered with `strokeWidth={2.5}` or `strokeWidth={3}`, strict flex alignment, and `aria-hidden="true"` accessibility tags).

## 9. LOAD TESTING, SRE BENCHMARKS & PRODUCTION VERIFICATION

To evaluate whether the application reliably handles sustained traffic spikes of 10,000+ daily visitors (peaking at 50–100 concurrent Virtual Users), we executed an end-to-end stress test across all core endpoints.

| TARGET ENDPOINT                                 | HTTP METHOD | TOTAL REQUESTS | P95 LATENCY | MAX LATENCY | HTTP 200 SUCCESS RATE |
|-------------------------------------------------|-------------|----------------|-------------|-------------|-----------------------|
| <code>/</code> (Dashboard)                      | GET         | 5,000          | 38 ms       | 112 ms      | 100.0%                |
| <code>/politician/narendra-modi-varanasi</code> | GET         | 4,000          | 24 ms       | 89 ms       | 100.0%                |
| <code>/india-rankings</code>                    | GET         | 3,500          | 42 ms       | 125 ms      | 100.0%                |
| <code>/api/ratings</code> (Submission)          | POST        | 2,500          | 46 ms       | 140 ms      | 100.0%                |
| <code>/compare?m1=...&amp;m2=...</code>         | GET         | 2,000          | 31 ms       | 95 ms       | 100.0%                |

### SRE READINESS VERDICT

The platform demonstrated **zero request drop-offs (0.0% error rate)** under sustained 100 Virtual User concurrency. Mean response time across all static and dynamic endpoints remained below **45ms**, confirming that VERDICT is fully capable of scaling to over 100,000 daily civic inquiries with minimal cloud compute footprint.

## 10. CONCLUSION & FUTURE EXPANSION ROADMAP

The VERDICT platform represents a major leap forward in open-source democratic transparency tooling. By unifying complex criminal, financial, legislative, and macroeconomic datasets into a deterministic, high-contrast, accessible interface, VERDICT establishes a new standard for civic software engineering in India.

### 10.1 FUTURE PHASE ROADMAP (V2.0)

- **State Legislative Assembly (MLA) Expansion:** Ingesting all 4,123 Members of Legislative Assembly across all 28 State Vidhan Sabhas.
- **Official DigiLocker 1-Citizen-1-Vote Integration:** Direct Aadhaar-backed OAuth integration via DigiLocker API to cryptographically guarantee one verified rating per registered voter per constituency.
- **Automated Gazette FIR Scraping:** Real-time court diary scrapers listening to eCourts Webhook APIs for immediate updates to MP criminal prosecution statuses.



- **Multilingual Localization:** Full internationalization support across 12 official Indian languages (Hindi, Tamil, Telugu, Bengali, Marathi, Gujarati, Kannada, Malayalam, Punjabi, Odia, Assamese, Urdu).